

GRC102 INFORMATION SECURITY GOVERNANCE

WEEK 4 PRACTICAL LABORATORY

Linux Security Monitoring, Auditing and Governance Assurance

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Assessment focus: auditd configuration and event evidence, Linux log analysis, Lynis security assessment, control monitoring, SIEM/automation mapping, remediation and retesting.

1. Executive Summary

This practical assessed Linux security monitoring and auditing controls in an authorised Ubuntu laboratory environment using auditd, journalctl, authentication logs and Lynis. The work focused on collecting technical evidence, interpreting security-relevant events, identifying control weaknesses and translating the results into governance, remediation and continuous-monitoring actions.

Auditd was operational and capable of recording security-relevant activity. Linux authentication and privileged-access logs provided attributable SSH and sudo activity. Lynis version 3.0.9 performed 257 tests and returned a hardening index of 60. Three explicit warnings were prioritised: security repository configuration requiring validation (PKGS-7388), vulnerable packages requiring remediation (PKGS-7392), and loaded iptables modules without active rules requiring validation of the effective firewall architecture (FIRE-4512).

The highest-priority condition is PKGS-7392 because known vulnerable packages may expose the system to exploitation if applicable updates are not assessed and remediated within approved timelines. Overall, the practical demonstrated that technical evidence becomes useful for governance only when it is linked to control objectives, ownership, escalation thresholds, remediation and retesting.

2. Scope and Authorisation

The assessment was conducted solely within the designated GRC102 Week 4 Ubuntu virtual machine. No destructive testing or unauthorised access activity was performed. The scope covered Linux audit configuration and event generation, system and authentication log review, Lynis-based security assessment, control-monitoring and governance mapping, and conceptual SIEM/continuous-monitoring integration.

3. Methodology

An evidence-based approach was used. Auditd service status and active rules were verified, benign activity was generated, and events were queried using ausearch and summarised with aureport. System and authentication logs were reviewed with journalctl and /var/log/auth.log for warnings, errors, SSH activity, failed-password indicators and sudo use. Lynis 3.0.9 was then used to assess the host; its report, log, warnings, suggestions and hardening index were retained. Finally, the technical results were mapped to control objectives, owners, monitoring thresholds, remediation actions, retest evidence and governance escalation.

4. Linux Log Analysis

The log review confirmed that authentication and privileged activity were available for audit review. No error-priority journal entries were returned during the reviewed period, and no confirmed failed-password attempt was identified. The failed-password search itself was visible in the sudo trail, which demonstrates why command context must be interpreted before classifying an event as an authentication failure.

Event / Condition	Evidence Source	User / Service	Security Significance	Recommended Action
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Successful SSH public-key authentication	/var/log/auth.log; journalctl	ubuntu / sshd	Remote authentication was logged and attributable.	Continue monitoring and investigate unexpected sources or accounts.
Privileged sudo activity	/var/log/auth.log	ubuntu / sudo	Provides an audit trail of privileged commands.	Monitor unusual or unauthorised sudo activity.
System warning events	journalctl -p warning	System/kernel services	May indicate configuration, availability or security conditions requiring review.	Review recurring warnings and escalate material conditions.
Error-priority review	journalctl -p err	System services	No error-priority entries returned during review period.	Continue periodic monitoring.
Failed-password review	/var/log/auth.log	Authentication services	No confirmed failed-password attempts identified.	Continue monitoring repeated failures or abnormal patterns.

5. Key Findings and Risk Priorities

ID	Finding	Priority	Owner	Risk / Significance	Recommended Action	Retest
W4-F01 / PKGS-7392	Vulnerable packages detected	High	Linux/System Administrator; Vulnerability Management	Known vulnerable software may increase exposure if affected packages remain unpatched.	Identify affected packages, assess applicable updates and apply approved remediation.	Repeat package vulnerability checks and rerun Lynis.
W4-F02 / PKGS-7388	Security repository not detected	High	Linux/System Administrator	Security-update source requires validation; incorrect configuration may affect timely patch delivery.	Validate APT sources and confirm the appropriate Ubuntu security repository is enabled and functioning.	Verify APT sources and repeat the relevant Lynis check.
W4-F03 / FIRE-4512	iptables modules loaded without active rules	Moderate	Linux/System Administrator; Security Operations	Effective host-level filtering requires validation. This warning alone does not establish absence of firewall protection.	Confirm intended firewall architecture and validate effective filtering controls.	Review effective firewall configuration and rerun Lynis firewall assessment.

6. Control-Monitoring Table

Control / Objective	Evidence	Owner	Status	Threshold	Risk	Remediation	Retest
Security auditing of sensitive activity	auditd enabled; custom rules	Linux/System Administrator	Operating	auditd and required rules remain active	Loss of audit coverage reduces accountability	Periodically verify auditd/rules	Confirm status/rules; query with ausearch
Authentication and privileged-access monitoring	auth.log SSH and sudo evidence	Linux/System Administrator / Security Operations	Operating	Activity remains logged; abnormal patterns investigated	Unauthorised access/privilege use may go undetected	Continue monitoring and alerting	Review auth.log/journal event capture
Vulnerability and patch management	Lynis PKGS-7392	Linux/System Administrator / Vulnerability Management	Requires Remediation	Vulnerable packages remediated within approved timelines	Known weaknesses may remain exploitable	Assess and apply approved updates	Repeat vulnerability checks and Lynis
Secure package-update configuration	Lynis PKGS-7388	Linux/System Administrator	Requires Validation	Approved security repositories configured and accessible	Updates may be delayed	Validate APT repository configuration	Verify sources and rerun Lynis

Host network filtering	Lynis FIRE-4512; firewall component active	Linux/System Administrator / Security Operations	Requires Validation	Filtering follows approved baseline	Unnecessary network exposure	Confirm intended architecture/effective filtering	Review ruleset and rerun Lynis
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7. Governance Escalation

- Highest priority: PKGS-7392 (vulnerable packages), because applicable known vulnerabilities should be assessed and remediated within approved risk and patch timelines.
- Ownership: vulnerable packages - Linux/System Administrator with Vulnerability Management support; security repository - Linux/System Administrator; firewall validation - Linux/System Administrator with Security Operations involvement.
- Escalation threshold: unresolved critical/high-risk vulnerabilities beyond approved timelines, repeated unexplained privileged access, loss of required audit logging, ineffective network controls, or overdue high-risk remediation.
- Closure evidence: updated package status, validated security repositories, effective firewall configuration, active audit rules and subsequent assessment results.
- Retesting: repeat the original test procedure after remediation wherever possible so before-and-after evidence can be compared.

8. SIEM and Continuous Monitoring Mapping

Evidence Source	Example Evidence	Monitoring / SIEM Use	Operational Alert	Governance Escalation
auditd	Sensitive-file activity, program execution, authentication-related audit events	Central collection, correlation and alerting	Unexpected sensitive-file change or suspicious execution	Required audit rules disabled, missing or repeatedly failing
Authentication / privileged logs	SSH authentication, sudo commands, failed logins, privileged sessions	Correlate access events and identify unusual patterns	Multiple failed logins or unexpected sudo activity	Persistent control failures, unexplained privileged access or inadequate logging
System logs	Service failures, warnings, kernel events and security-relevant errors	Centralise recurring errors and abnormal service behaviour	Security-critical service failure or unusual event	Repeated control-related failures or prolonged logging outage
Lynis	Vulnerable packages, repository warning, firewall warning, hardening suggestions	Vulnerability/configuration monitoring and remediation dashboards	New high-risk technical weakness	High-risk findings overdue or repeatedly reappearing

9. Remediation and Retest Plan

Finding	Priority	Owner	Remediation	Retest Evidence	Escalation Condition
PKGS-7392	High	Linux/System Administrator / Vulnerability Management	Identify affected packages; assess and apply approved updates.	Package status and Lynis rerun.	Escalate if remediation exceeds approved SLA/risk tolerance.
PKGS-7388	High	Linux/System Administrator	Validate configured APT repositories and security-update source.	APT source validation and relevant Lynis rerun.	Escalate if security updates cannot be received within approved timeframe.
FIRE-4512	Moderate	Linux/System Administrator / Security Operations	Validate intended firewall framework and effective filtering.	Effective ruleset review and Lynis rerun.	Escalate if required filtering is absent or ineffective.

The complete submission package contains nine screenshot exhibits across the Auditd, Log Analysis and Lynis evidence folders.

Appendix C - AI Assistance Declaration

AI assistance was used for report structuring and language refinement. The laboratory activities, commands, evidence collection, technical verification and review of findings were performed by the student in the authorised lab environment.